

over 10,000 hackers and a capability second only to that of the US. They are organised under the "Fourth Division" of the PLAN and virus-writing is taught in Chinese military schools. Backing up this "informationalised army" is The Red Hackers' Alliance, the fifth largest group of hackers in the world which is believed to help the Chinese government in executing global cyber attacks. This is a growing capability that has worried India. "You must have heard what China is doing (in cyber warfare) and there is nothing to stop them unless we keep developing our prowess in cyber warfare, together with other ingredients of comprehensive national power," NTRO Chairman KVSS Prasad Rao said in a talk at the United Service Institution of India just two months before the cyber break-in.

The December 15 attack was more of a cyber-espionage mission to ferret out secrets from top government officials rather than to disable computers. It came in the form of an innocuous pdf file titled "Save Ganga Mission". It was sent from a defunct .nic.in address of an official with an embedded Trojan Horse programme which, like the mythical Greek military ruse, spirited a "malware" into computers. The Trojan would turn the infected computer into a "zombie", allowing a hacker to remotely control the machine and capture its data. In Delhi, the investigators immediately began a frantic mission to track the source of the attack. The Trojan was "sand-bagged" just as a bomb disposal team would isolate an explosive device. The e-mail was traced to California's San Jose State University. Here, investigators found the servers had been hacked by the Russian Business Network (RBN), an euphemism for a cyber crime hosting network based in Russia. Hacking into RBN, they detected Chinese signatures but before they could expose the main channel, the trail went cold. It was the first time that Indian government operatives had engaged in a deep penetration operation. It took a small group of Indian cyber warriors 10 sleepless days and nights to come

Cyber Weapons

These are software codes that can be spread across a network. They are easy to replicate and disperse in large numbers but difficult to trace back to the source.

Malware: Short for malicious software, designed to infiltrate a computer system without the user's informed consent. Includes viruses and worms.

Worms: A self-replicating programme that affects the infected computer by overloading.

Spyware: A malware that lodges itself in a user's computer and tracks computer usage patterns.

Key Loggers: A programme covertly installed on a computer to track the usage of the keyboard.

Trojan Horses: A kind of non-self-replicating malware when planted on a user's computer. It can be remotely accessed by a hacker's computer. Computers infected by Trojans can be used for distributed denial of services or DDos attacks.

Bots: A software application that does repeated tasks on the Internet. The most common Internet bots are the spider bots used for web server analyses and file data gathering.

BotNets: A collection of compromised computers or zombie computers running software usually installed by downloads. BotNets operate under a common command-and-control infrastructure operated by a hacker.

SOUTH BLOCK HAS SEEN WEB-BASED ATTACKS BY CHINESE HACKERS



this far and trace the origin of the attack. In January, an identical attack was reported against computers of the National Security Advisory Board and the National Security Council. Trojans were sent in an attachment titled 'National Security Document' to 200 computers.

So far the Indian Government policy has been defensive. The Computer Emergency Response Team (CERT) within the Department of Information Technology is meant to defend India's IT infrastructure. It acts in coordination with the National Informatics Centre, the government's service provider. Information warfare groups of the DIA and NTRO are the only ones tasked with offensive cyber warfare. India spends less than \$1 million (Rs 5 crore) on offensive cyber warfare and has only a handful of experts. The Chinese, in sharp contrast, have an annual budget of \$55 million and employ thousands of hackers.

The operations of this hacking army were revealed in a 10-month investigation by Greg Walton and Shishir Nagaraja in March last year. Working for the Canada-based Information Warfare Monitor, the team uncovered GhostNet, a massive global Chinese information gathering